

AS Chemistry homework · 3.1 Electronegativity

CIE 9701 AS Chemistry · 16 marks · about 35–40 minutes

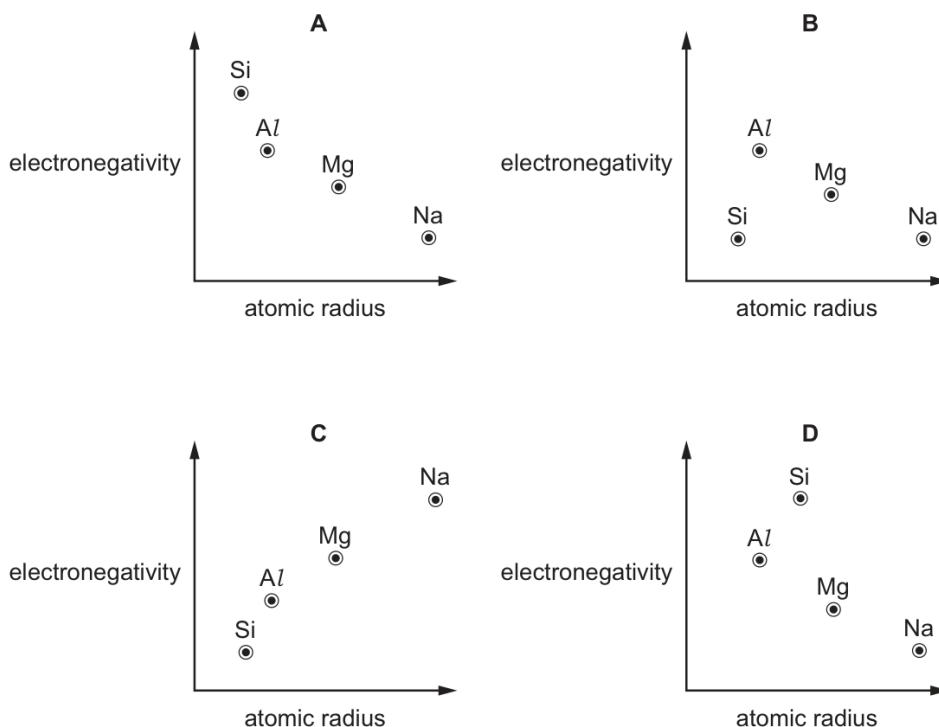
Topic: 3.1 Electronegativity and bonding — definition, factors, Periodic Table trends, and using Pauling differences to predict ionic or covalent bonding. Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; give the usual CIE reasons (nuclear charge, atomic radius, shielding). Pauling values are given on the paper where needed.

Section	Content	Marks
A	Multiple choice	11
B1	2023 FM Paper 22 Question 1(a)	3
B2	2023 MJ Paper 22 Question 1(b)(ii)	2
Total		16

Section A — Multiple-choice questions (11 marks)**A1.**

[1]

- 19 Which graph correctly shows relative electronegativity plotted against relative atomic radius for the elements Na, Mg, Al and Si?



2022 MJ Paper 12 Question 19

A2.

[1]

V and Z are both elements in Period 3 of the Periodic Table. Each element forms one stable ion that does not contain another element.

The atomic radius of each element and the ionic radius of the ion described above is shown.

element	atomic radius / nm	ionic radius / nm
V	0.186	0.095
Z	0.099	0.181

Which statement is correct?

- A** Ions of V and Z have the same number of full electron shells.
- B** Ions of Z are positively charged.
- C** Z has a greater electronegativity than V.
- D** V has more outer electrons than Z.

2023 FM Paper 12 Question 22

A3.

[1]

The elements in Period 3 and their compounds show trends across the period from sodium to chlorine.

Which row is correct?

	electronegativity of the elements	acid/base behaviour of the oxides of the elements
A	decreases	basic → amphoteric → acidic
B	decreases	acidic → amphoteric → basic
C	increases	basic → amphoteric → acidic
D	increases	acidic → amphoteric → basic

2022 FM Paper 12 Question 18

A4.

[1]

Which row shows the expected properties of the element astatine when compared to the properties of the element iodine?

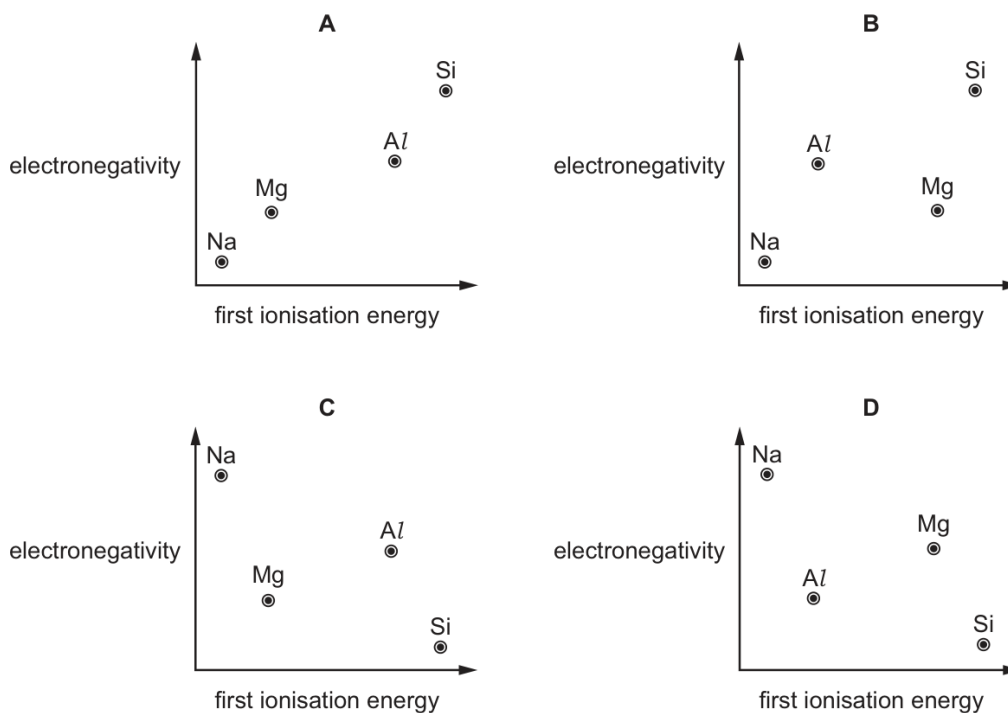
	electronegativity of astatine compared to iodine	volatility of astatine compared to iodine
A	less electronegative	higher
B	more electronegative	higher
C	less electronegative	lower
D	more electronegative	lower

2023 MJ Paper 13 Question 17

A5.

[1]

20 Which diagram shows the electronegativity of the elements Na, Mg, Al and Si plotted against their first ionisation energies?

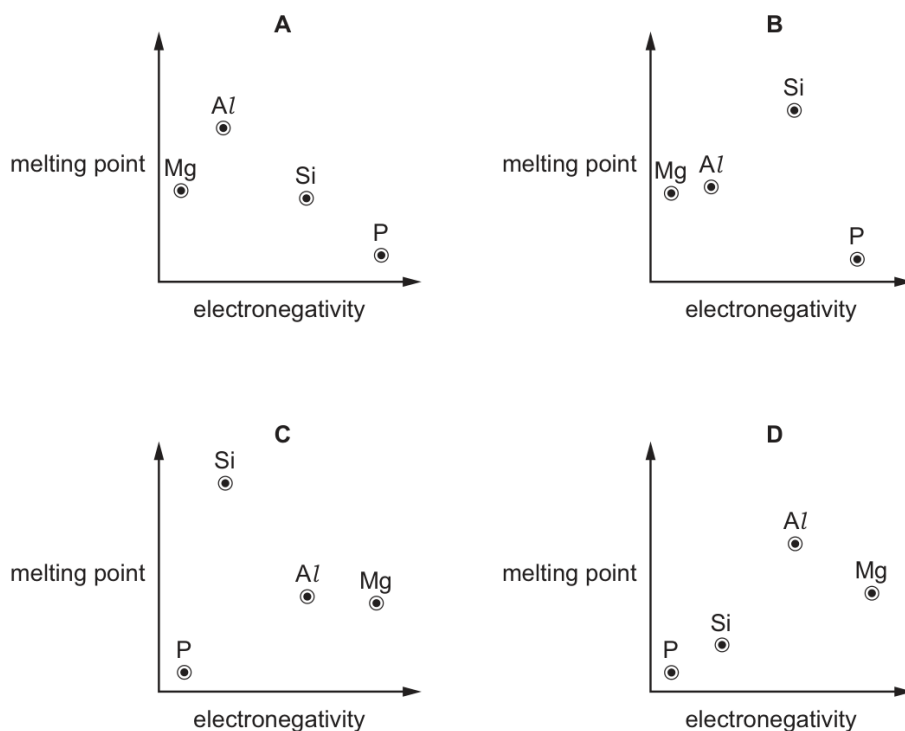


2025 MJ Paper 11 Question 20

A6.

[1]

- 17 Which graph shows the relative melting points of the elements Mg, Al, Si and P plotted against their relative electronegativities?



2025 ON Paper 14 Question 17

A7.

[1]

Which compound has the smallest difference in electronegativity between its two elements?

A KF **B** KBr **C** LiF **D** LiBr

2022 MJ Paper 13 Question 5

A8.

[1]

L and M are elements in Period 3 of the Periodic Table. Neither element is argon.

Information about the Pauling electronegativity values of L and M is given.

element	Pauling electronegativity value
L	the highest of the seven elements Na to Cl
M	the lowest of the seven elements Na to Cl

Three statements about elements L and M are given.

1. Element L contains covalent bonds.
2. Element L has a higher atomic number than element M.
3. A compound of L and M contains ionic bonds.

Which statements are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

2023 MJ Paper 13 Question 4

A9.

[1]

Elements X, Y and Z are all in the first two periods of the Periodic Table.

Their Pauling electronegativity values, E_N , are shown.

element	E_N
X	1.0
Y	2.1
Z	4.0

Substances exist with formulae XZ, YZ and Z_2 .

Which row puts these substances in order of increasing melting point?

	lowest melting point	→	highest melting point
A	XZ	YZ	Z_2
B	XZ	Z_2	YZ
C	Z_2	YZ	XZ
D	Z_2	XZ	YZ

2022 FM Paper 12 Question 7

A10.

[1]

Which type of bonding is **never** found in elements?

A covalent**B** ionic**C** metallic**D** van der Waals' forces

2021 MJ Paper 13 Question 6

A11.

[1]

Sodium, magnesium, aluminium, silicon and phosphorus are all elements in Period 3 of the Periodic Table.

Three statements about the oxides and chlorides of these elements are given.

1. The ionically bonded oxides all react with dilute hydrochloric acid.
2. All metal chlorides produce neutral solutions when added to water.
3. The two most electronegative elements both form covalently bonded chlorides.

Which statements are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

2023 MJ Paper 12 Question 19

Section B — Structured questions (5 marks)

B1 · 2023 February/March Paper 22 Question 1(a) (3 marks)

The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6

Table 1.1

B1(a)(i)

[1]

Define electronegativity.

B1(a)(ii)

[2]

O and S are in Group 16.

Explain the difference in the Pauling electronegativity values of O and S.

B2 · 2023 May/June Paper 22 Question 1(b)(ii) (2 marks)

B2(b)(ii)

[2]

Explain why electronegativity increases across a period.